Mathematical Statistics

Lecture 09 Continuous Random Variables & Probability Density Function

Zamir Hussain
University of Wah

November 11, 2025

Contents

1 Lecture Overview

This lecture transitions from discrete random variables to **continuous random variables**. We will define what a continuous random variable is and introduce its counterpart to the probability mass function: the **Probability Density Function (PDF)**. We will explore the fundamental properties of a PDF, learn how to use it to calculate probabilities, and understand its relationship with the **Cumulative Distribution Function (CDF)**. Several worked examples will be provided to solidify these concepts.

2 Continuous Random Variables

2.1 Definition

A **continuous random variable** is a variable that can take on an **infinite** number of possible values within a given interval. Unlike discrete variables, which have countable values (e.g., the roll of a die), continuous variables can assume any value in a range.

From Summation to Integration

With discrete variables, we sum up probabilities using $\sum$. With continuous variables, there are infinitely many points in any interval, so the probability of hitting one exact point is zero. Instead, we find the probability over an interval by measuring the area under a curve. This is why we shift from summation to **integration** ($\int$).

2.2 Examples

- **Height of a person:** A person's height isn't restricted to 175 cm or 176 cm; it can be 175.1 cm, 175.11 cm, and so on.
- **Weight of an object:** Similar to height, weight can take any value within a range.
- **Time:** The time it takes to complete a task.
- **Temperature:** The temperature of a room.

3 The Probability Density Function (PDF)

For a continuous random variable, the probability of it taking on any *single, specific* value is zero. For example, $P(\text{Height} = 175.000...\text{cm}) = 0$. Instead, we talk about the probability that the variable falls within an **interval**. This is described by the **Probability Density Function (PDF)**, denoted as $f(x)$.

3.1 Core Properties of a PDF

Core Properties of a PDF

For a function $f(x)$ to be a valid PDF, it must satisfy two essential conditions:

1. **Non-negativity:** The function must be greater than or equal to zero for all possible values of x.
$$f(x) \geq 0 \quad \text{for all } x$$
2. **Total Area is One:** The total area under the curve of the function over its entire domain ($-\infty$ to ∞) must be equal to 1. This is the continuous equivalent of the sum of all probabilities being 1 for discrete variables.
$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

3.2 Calculating Probability over an Interval

The probability that a continuous random variable X falls between two values, a and b, is the area under the PDF curve from a to b.

$$P(a \leq X \leq b) = \int_a^b f(x)\,dx$$

4 Worked Examples

4.1 Example 1: Verifying a PDF

Problem: Show that the function $f(x)$ is a valid PDF.

$$f(x) = \begin{cases} x, & \text{if } 0 \leq x < 1 \\ 2 - x, & \text{if } 1 \leq x < 2 \\ 0, & \text{otherwise} \end{cases}$$

Solution: To verify that $f(x)$ is a PDF, we must check if its total integral equals 1.

1. **Set up the integral:** We need to compute $\int_{-\infty}^{\infty} f(x)\,dx$.
2. **Break the integral into defined pieces:** Since $f(x)$ is zero outside the interval $[0, 2]$, we only need to integrate over that range.
$$\begin{aligned}\int_{-\infty}^{\infty} f(x)\,dx &= \int_{-\infty}^{0} 0\,dx + \int_0^1 x\,dx + \int_1^2 (2-x)\,dx + \int_2^{\infty} 0\,dx \\ &= 0 + \int_0^1 x\,dx + \int_1^2 (2-x)\,dx + 0\end{aligned}$$
3. **Evaluate the non-zero integrals:**
   - $\int_0^1 x\,dx = \left[\frac{x^2}{2}\right]_0^1 = \frac{1^2}{2} - \frac{0^2}{2} = \frac{1}{2}$
   - $\int_1^2 (2-x)\,dx = \left[2x - \frac{x^2}{2}\right]_1^2 = \left(2(2) - \frac{2^2}{2}\right) - \left(2(1) - \frac{1^2}{2}\right) = (4-2) - (2 - \frac{1}{2}) = 2 - \frac{3}{2} = \frac{1}{2}$

4. **Sum the results:**

$$\text{Total Integral} = \frac{1}{2} + \frac{1}{2} = 1$$

Conclusion: Since the total integral is 1 and the function is non-negative on its domain, $f(x)$ is a valid Probability Density Function.

5 The Cumulative Distribution Function (CDF)

The **Cumulative Distribution Function (CDF)**, denoted as $\boldsymbol{F(x)}$, gives the total probability that the random variable X is less than or equal to a specific value x.

5.1 Definition and Formula

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)\, dt$$

(Note: We use a different variable, t, for integration to avoid confusion.)

5.2 Key Relationship between PDF and CDF

The Fundamental Theorem of Probability

The PDF is the derivative of the CDF. This is a fundamental relationship in probability theory, analogous to the Fundamental Theorem of Calculus.

$$f(x) = \frac{d}{dx}[F(x)] = F'(x)$$

This means if you have the CDF, you can find the PDF by differentiating it.

5.3 Example 2: Finding the CDF

Problem: Find the CDF, $F(x)$, for the PDF from Example 1.
Solution: We calculate $F(x) = \int_{-\infty}^{x} f(t)\, dt$ for each interval.

- **Case 1:** $x < 0$
$$F(x) = \int_{-\infty}^{x} 0\, dt = 0$$
- **Case 2:** $0 \leq x < 1$
$$F(x) = \int_{-\infty}^{0} 0\, dt + \int_{0}^{x} t\, dt = 0 + \left[\frac{t^2}{2}\right]_0^x = \frac{x^2}{2}$$
- **Case 3:** $1 \leq x < 2$ Here, we accumulate the probability up to 1 and then add the integral from 1 to x.
$$\begin{aligned} F(x) &= P(X \leq 1) + \int_{1}^{x} (2-t)\, dt \\ &= \frac{1^2}{2} + \left[2t - \frac{t^2}{2}\right]_1^x \\ &= \frac{1}{2} + \left(2x - \frac{x^2}{2}\right) - \left(2(1) - \frac{1^2}{2}\right) \\ &= \frac{1}{2} + 2x - \frac{x^2}{2} - \frac{3}{2} = 2x - \frac{x^2}{2} - 1 \end{aligned}$$
- **Case 4:** $x \geq 2$ The total accumulated probability up to 2 (and beyond) is 1. So, $F(x) = 1$.

Final CDF: The complete CDF, $F(x)$, is the piecewise function:

$$F(x) = \begin{cases} 0, & \text{if } x < 0 \\ \frac{x^2}{2}, & \text{if } 0 \leq x < 1 \\ 2x - \frac{x^2}{2} - 1, & \text{if } 1 \leq x < 2 \\ 1, & \text{if } x \geq 2 \end{cases}$$

6 Practical Applications in Data Science & ML

Understanding Your Data's Distribution

In Data Science and Machine Learning, you rarely work with perfect mathematical functions. Instead, you work with real-world data. Understanding the underlying probability distribution of your data (your features) is a fundamental skill.

- **Model Assumptions:** Many ML models, like Linear Regression, assume that the errors are normally distributed. Validating this assumption is crucial for model reliability.
- **Anomaly Detection:** If you know the PDF of "normal" behavior (e.g., server response times), you can identify an anomaly as a data point with a very low probability density. This means it's a rare event that warrants investigation.
- **Feature Engineering:** Transforming a feature to have a more "well-behaved" distribution (like a normal distribution) can significantly improve the performance of some models.

The bridge between the theoretical PDF we study in class and real data is the **histogram**. A histogram shows the empirical distribution of the data, and we can fit a theoretical PDF to it to model the underlying process.

6.1 Practical Assignment: Analyzing Simulated Student Test Scores

Python Assignment: From Data to PDF

Objective: To simulate a dataset of student test scores, visualize its distribution using a histogram, fit a theoretical Normal Distribution PDF to it, and use the model to calculate probabilities.
Tools: You will need Python with the following libraries: 'numpy', 'matplotlib', and 'scipy'.

Step 1: Generate Simulated Data
Let's assume student test scores follow a Normal (Gaussian) distribution. We will generate 1000 sample scores with a mean (μ) of 75 and a standard deviation (σ) of 10.

</> Python Code

```
import numpy as np

# Parameters for our distribution
mu, sigma = 75, 10
# Generate 1000 data points
scores = np.random.normal(mu, sigma, 1000)
```

Step 2: Visualize the Data with a Histogram
A histogram groups the data into bins and counts how many data points fall into each bin. By setting 'density=True', the histogram is normalized so that the total area of the bars equals 1, making it a direct visual analogue of a PDF.

</> Python Code

```
import matplotlib.pyplot as plt

plt.figure(figsize=(10, 6))
# Plot the histogram
plt.hist(scores, bins=30, density=True, alpha=0.6, color='g',
    label='Empirical Data (Histogram)')
plt.title('Distribution of Student Test Scores')
plt.xlabel('Score')
plt.ylabel('Density')
plt.legend()
plt.grid(True)
plt.show()
```

Step 3: Fit a Theoretical PDF
Now, let's overlay the theoretical PDF of the Normal distribution using the same mean and standard deviation. We use 'scipy.stats.norm' for this. This shows how well our theoretical model fits the empirical data.

Python Code

```
from scipy.stats import norm

# Generate a range of x values for the PDF plot
x_axis = np.linspace(mu - 4*sigma, mu + 4*sigma, 1000)
# Calculate the PDF for these x values
pdf_values = norm.pdf(x_axis, mu, sigma)

# Plot the histogram AND the PDF
plt.figure(figsize=(10, 6))
plt.hist(scores, bins=30, density=True, alpha=0.6, color='g',
    label='Empirical Data (Histogram)')
plt.plot(x_axis, pdf_values, 'r-', lw=2, label='Theoretical PDF
    (Normal Dist.)')
plt.title('Distribution of Student Test Scores')
plt.xlabel('Score')
plt.ylabel('Density')
plt.legend()
plt.grid(True)
plt.show()
```

Step 4: Calculate Probabilities using the CDF
Let's answer a practical question: What is the probability that a randomly selected student scored between 60 and 80? In class, you'd solve this with an integral: $\int_{60}^{80} f(x)\,dx$. Computationally, we use the Cumulative Distribution Function (CDF), where $P(a < X < b) = F(b) - F(a)$.

Python Code

```
# Calculate P(60 < score < 80)
# F(80) - F(60)
prob_60_to_80 = norm.cdf(80, mu, sigma) - norm.cdf(60, mu, sigma)

print(f"The probability of a score between 60 and 80 is:
    {prob_60_to_80:.4f}")

# Expected output:
# The probability of a score between 60 and 80 is: 0.6247
```

Challenge Question: What is the score that 90% of students scored below? (This is the 90th percentile). *Hint: You need the inverse of the CDF, which is called the Percent Point Function ('ppf').*